# MILITARY SPECIFICATION

## ARM-SAFE DEVICE: M120

### 1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of Arm Safe Device (ASD), M120 which is used with the Lance warhead section.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-202 | - Test Methods for Electronic and Electrical Component Parts. |
| MIL-STD-1169 | - Packaging, Packing and Marking for Shipment of Inert Ammunition Components. |
| MIL-STD-1460 | - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For. |
| MIL-STD-20056 | - Insert Arrangement, Electric Connector, Shell Size 23. |

DRAWINGS

PICATINNY ARSENAL

| | |
|---|---|
| 8880002 | - Arm/Safe Device, AW:M120. |
| EL-8880002 | - Equipment List. |
| 8880086 | - Base Assembly. |

PUBLICATIONS

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

NUOR

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The ASD shall comply with all requirements specified on the drawing (dwg) 8880002, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Circuit resistance. The resistance of the independent circuits listed shall not exceed 55 milliohms. The circuit resistance of A to a shall not exceed 75 milliohms when tested as specified in 4.4.1. The circuit resistance for each of the three independent monitor switches will be 55 milliohms maximum. Table I lists the terminals corresponding to switches for each accelerometer (identified by location of Figure 1). Table II lists the circuits to be tested.



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Table I

Switch-Pin Correlation

| Accelerometer No. 1 | | Test Code No. (See 3.17) | Accelerometer No. 2 | | Test Code No. (See 3.17) | Accelerometer No. 3 | | Test Code No. (See 3.17) |
|---|---|---|---|---|---|---|---|---|
| Switch | Pins | | Switch | Pins | | Switch | Pins | |
| S1-M | A-B | 050001 | S2-M | B-W | 050015 | S3-M | W-a | 050029 |
| S1-1 | C/D-d | 050002 | S2-1 | d-X/Y | 050016 | S3-1 | X/Y-y | 050030 |
| S1-2 | C/D-y | 050003 | S2-2 | x-C/D | 050017 | S3-2 | X/Y-x | 050031 |
| S1-3 | V/c-e | 050004 | S2-3 | e-M/N | 050018 | S3-3 | M/N-w | 050032 |
| S1-4 | V/c-w | 050005 | S2-4 | v-V/c | 050019 | S3-4 | M/N-v | 050033 |
| S1-5 | R/S-f | 050006 | S2-5 | f-J/K | 050020 | S3-5 | J/K-u | 050034 |
| S1-6 | R/S-u | 050007 | S2-6 | t-R/S | 050021 | S3-6 | J/K-t | 050035 |
| S1-7 | G/H-g | 050008 | S2-7 | s-T/U | 050022 | S3-7 | T/U-s | 050036 |
| S1-8 | G/H-s | 050009 | S2-8 | r-G/H | 050023 | S3-8 | T/U-r | 050037 |
| S1-9 | E/F-h | 050010 | S2-9 | h-Z/b | 050024 | S3-9 | Z/b-q | 050038 |
| S1-10 | E/F-q | 050011 | S2-10 | p-E/F | 050025 | S3-10 | Z/b-p | 050039 |
| S1-11 | AA/BB-k | 050012 | S2-11 | k-CC/DD | 050026 | S3-11 | CC/DD-n | 050040 |
| S1-12 | AA/BB-n | 050013 | S2-12 | m-AA/BB | 050027 | S3-12 | CC/DD-m | 050041 |



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Table II

Circuit Resistance

| Accelerometer Condition | | | Circuits to be Tested | |
|---|---|---|---|---|
| S1 | S2 | S3 | | |
| Armed | Safe | Safe | Switches: S1-1 through S1-12 | (See |
| Safe | Armed | Safe | Switches: S2-1 through S2-12 | Table I |
| Safe | Safe | Armed | Switches: S3-1 through S3-12 | for corres. |
| Safe | Safe | Safe | Switches: S1-M, S2-M and S3-M terminals) | |

| Terminals: | Test Code No. (See 3.17) |
|---|---|
| C to D | 050043 |
| E to F | 050044 |
| G to H | 050045 |
| J to K | 050046 |
| M to N | 050047 |
| R to S | 050048 |
| T to U | 050049 |
| V to c | 050050 |
| X to Y | 050051 |
| Z to b | 050052 |
| AA to BB | 050053 |
| CC to DD | 050054 |
| A to a | 050055 |

Table III

Insulation Resistance

| ASD Condition | Terminals | Test Code No. (See 3.17) | Terminals | Test Code No. (See 3.17) | Test |
|---|---|---|---|---|---|
| Safe | A | | h | | Each pin |
| | C | | k | | to all |
| | E | | m | | other |
| | G | | n | | listed |
| | J | | p | | pins and |
| | M | | q | | case |
| | R | | r | | tied |
| | T | | s | | together |
| | V | | t | | |
| | X | | u | | |
| | Z | | v | | |
| | d | | w | | |
| | e | | x | | |
| | f | | y | | |
| | g | | AA | | |
| | | | CC | | |
| | EE,FF,GG,HH,L,P and z tied together as one terminal | | | | |



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| | | | |
|---|---|---|---|
| Armed | A | J | Each pin |
| | B | M | to all |
| | C | W | other |
| | E | a | listed |
| | G | AA | pins and |
| | | | case tied |
| | EE,FF,GG,HH,L,P and z tied together as one terminal | | together |

3.7 Insulation resistance. The insulation resistance of each circuit shall be 50 megohms minimum at 500 ± 50 volts direct current (vdc) in both the safe and armed positions when tested as specified in 4.4.2. Table III lists the terminals to checked.

3.8 Seal integrity. The ASD shall be subjected to a water immersion type bubble leak test. When tested as specified in 4.4.3, no leakage bubble shall be observed.

3.9 Operation.

3.9.1 Safing at low temperature (-40°F). The normally open circuits shall not close when 11 g's are applied for 10 or more seconds, when tested as specified in 4.4.4.1. While at this temperature, the unit shall comply with the requirements of 3.6 and 3.7.

3.9.2 Arming at low temperature (-40°F). The normally open functional switches shall be closed and latched after the application of a nominal 14 g-second product. Upon application of 14 g's, the functional switches shall operate within 1 ± 0.2 seconds when tested as specified in 4.4.4.1. Upon completion of this test, and while at this temperature, the unit shall meet the requirements of 3.6 and 3.7.

3.10 Operation.

3.10.1 Safing. The normally open circuits shall not close when 11 g's are applied for 10 or more seconds, when tested as specified in 4.4.5.1. The monitor switches shall close when the 11 g field is removed.

3.10.2.1 Arming at low temperature (-40°F). The normally open functional switches shall be closed and latched after the application of nominal 14 g-second product. Upon application of 14 g's, the functional switches shall operate within 1 ± 0.2 seconds when tested as specified in 4.4.5.2.

| Functional Switch No. | Test Code No. |
|---|---|
| S1 | 092101 |
| S2 | 092102 |
| S3 | 092103 |



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3.10.2.2 Arming at high temperature (+160°F). The normally open functional switches shall be closed and latched after the application of a nominal 14 g-second product. Upon application of 14 g's, the functional switches shall operate within 1 ± 0.08 seconds when tested as specified in 4.4.5.4. Upon completion of this test, the unit shall meet the requirements of 3.6 and 3.7.

| Functional Switch No. | Test Code No. |
|---|---|
| S1 | 092201 |
| S2 | 092202 |
| S3 | 092203. |

3.10.2.3 Arming at ambient temperature (+80°F). The normally open functional switches shall be closed and latched after the application of a nominal 14 g-second product. Upon application of 14 g's, the functional switches shall operate within 1 + 0.08 seconds when tested as specified in 4.4.5.4. Upon completion of this test, the unit shall meet the requirements of 3.6 and 3.7.

| Functional Switch No. | Test Code No. |
|---|---|
| S1 | 092301 |
| S2 | 092302 |
| S3 | 092303. |

3.10.2.4 If one (1) of the three (3) Auxiliary Switches fails to function, a determination must be made to assure and verify that each of the 3 Actuator Switch Assemblies has funtioned properly. If in this case, the 3 Actuation Switch Assemblies are found to have functioned properly, then the ASD will be considered to have successfuly passed its operational test. If 2 or more of the 3 Auxiliary Switches fail to function, the ASD will be considered to have failed its operational test.

3.11 Mechanical shock. The device in the safe condition, after being stabilized at -65°F, or +160°F as applicable, shall be subjected to 3 shocks in each direction along each of 3 mutually perpendicular axis (for a total of 18 shocks) (see Figure 1) as specified in 4.4.6, according to Table IV. There shall be no evidence of damage (see 6.5).



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TABLE IV

SHOCK, MECHANICAL

| ASD Condition | Temperature Tol + 5°F | Shock Peak Magnitude | Shock Duration (Tol + 1 ms) | Shock Pulse Shape |
|---|---|---|---|---|
| Safe | -65°F | 50 g | 11 ms | half sine |
| Safe | +160°F | " | " | " " |

3.12 Operational vibration. The ASD, after being stabilized at -40°F or +160°F as applicable and in the armed condition, shall be subjected to vibration for the times and at the temperatures specified in Table V in each of three mutually perpendicular axes (see Figure 1) without evidence of damage (see 6.5). The instantaneous peak input amplitude shall be limited to 3 times the rms acceleration level. Testing shall be as specified in 4.4.7. There shall be no evidence of damage (see 6.5).

TABLE V

OPERATIONAL VIBRATION

SPECTRUM - 1

| Frequency Ranges,(Hz) | Power Spectral Density ($g^2$/Hz) or Attenuation (db/octave) | Equalization |
|---|---|---|
| Below 50 | +18db/octave | Min slope |
| 50-100 | +6db/octave | +3db |
| 100-1000 | 0.06$g^2$/Hz | ±3db-1.5db |
| 1000-2000 | -6db octave | +3db |
| Above2000 | -18db/octave | Min slope |

rms overall 9.3g's Minimum

OPERATING CONDITIONS - 1

| Cond | Time per Axis | Temp (°F) |
|---|---|---|
| Armed | 10sec± 0.5sec | -40°F±5°F |
| Armed | 10sec± 0.5sec | +160°F±5°F |

SPECTRUM - 2

| Frequency Ranges,(Hz) | Power Spectral Density ($g^2$/Hz) or Attenuation (db/octave) | Equalization |
|---|---|---|
| Below 50 | +18db/octave | Min slope |
| 50-100 | +6db/octave | +3db |
| 100-1000 | 0.02$g^2$/Hz | +3db,-1.5db |
| 1000-2000 | -6db/octave | +3db |
| Above2000 | -18db/octave | Min slope |

rms overall 5.3g's Minimum

OPERATING CONDITIONS - 2

| Cond | Time per Axi | Temp(°F) |
|---|---|---|
| Armed | 4.5min±14sec | +160°F±5°F |
| Armed | 4.5min±14sec | -40°±5°F |



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3.13 Transportation vibration. The ASD, after being stablized at -65°F or +160°F as applicable and in the safe condition (as mounted), shall be vibrated at frequencies and magnitudes (gravity units) (g's), or inches double amplitude) listed in Table VI, sweeping the entire frequency band at a logarithmic rate such that a complete sweep (5-500-5) requires 15 minutes, in each of three mutually perpendicular axes (see Figure 1), without evidence of damage (see 6.5) when tested as specified in 4.4.8.

Table VI

Transportation Vibration

| Frequency, Hz | Magnitude | Temperature (Tol + 5°F) | Sweeps Per Axis |
|---|---|---|---|
| 5 - 15-5 | 0.4 in. da | -65°F | 1 |
| 15 - 80 | 5.0 g | Ambient | 3 |
| 80 - 115 | 0.015 in. da | +160°F | 1 |
| 115 - 500 | 10.0 g | | |

3.14 Temperature shock. The device in the safe condition shall withstand sudden changes in temperature without evidence of damage (see 6.5) when tested as specified in 4.4.9.

3.15 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.16 Reliability and probability of compliance. (Information concerning the reliability and probability of compliance will be supplied following completion of Qualification and Reliability Tests).

3.17 Test identification number. The contractor shall provide variables reading on punch cards using the six digit number designated in the applicable paragraph in Section 3. Punch cards shall be prepared and mailed in accordance with QAI-PA-1.

3.18 Workmanship.

3.18.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.18.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.18.3 Burrs. Parts shall be free from burrs.

3.18.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.18.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 8 complete assemblies (see dwg 8880002) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VII. Tests shall be conducted in order specified in Table VII.

Table VII

First article inspections and tests

| Inspections and Tests | Reqmt Para. | Item Numbers 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | Dest | Non-Dest |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Classification of Characteristics | 4.3.2.1 | X | X | X | X | X | X | X | X | | X |
| Circuit resistance | 3.6 | X | X | X | X | X | X | X | X | | X |
| Insulation resistance | 3.7 | X | X | X | X | X | X | X | X | | X |
| Seal | 3.8 | X | X | X | X | X | X | X | X | | X |
| Operation | 3.9 | X | X | X | X | X | X | X | X | | X |
| Temperature shock | 3.14 | | X | X | X | X | | | | X | |
| Transportation vibration 160°F | 3.13 | | X | X | | | X | X | | X | |
| Transportation vibration -65°F | 3.13 | | X | | X | | X | | X | X | |
| Mechanical shock 160°F | 3.11 | | X | | | X | | X | X | X | |
| Mechanical shock -65°F | 3.11 | | | X | X | | | X | X | X | |
| Operation -40°F | 3.9 | X | X | X | X | X | X | X | X | | X |
| Operational vibration +160°F | 3.12 | | | X | | X | X | | X | X | |
| Operational vibration-40°F | 3.12 | | | | X | X | X | X | | X | |
| Circuit resistance (Armed) | 3.6 | X | X | X | X | X | X | X | X | | X |
| Insulation resistance (Armed) | 3.7 | X | X | X | X | X | X | X | X | | X |
| Seal | 3.8 | X | X | X | X | X | X | X | X | | X |



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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. The formation of an inspection lot and related functions shall be in accordance with Section 5, MIL-STD-105. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VIII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristics inspections are performed.

TABLE VIII

| Major Characteristics | | Minor Characteristics | |
|---|---|---|---|
| Lot Size | Sample Size | Lot Size | Sample Size |
| 1-155 | 100% inspection | 1-25 | 100% inspection |
| 156-200 | 155 | 26-30 | 25 |
| 201-250 | 175 | 31-35 | 26 |
| 251-300 | 190 | 36-30 | 27 |
| 301-350 | 200 | 41-45 | 28 |
| 351-400 | 210 | 46-50 | 29 |
| 401-450 | 220 | 51-55 | 30 |
| 451-500 | 225 | 56-60 | 31 |
| 501-550 | 230 | 61-65 | 32 |
| 551-600 | 235 | 66-70 | 33 |
| 601-650 | 240 | 71-75 | 34 |
| 651-700 | 243 | 76-85 | 35 |
| 701-750 | 247 | 86-95 | 36 |
| 751-800 | 250 | 96-110 | 37 |
| 801-900 | 254 | 111-130 | 38 |
| 901-1000 | 258 | 131-156 | 39 |



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| | | | |
|---|---|---|---|
| 1001-1100 | 261 | 157-190 | 40 |
| 1101-1200 | 264 | 191-240 | 41 |
| 1201-1300 | 266 | 241-340 | 42 |
| 1301-1400 | 268 | 341-520 | 43 |
| 1401-1500 | 270 | 521-1120 | 44 |
| 1501-1600 | 272 | 1121 and over | 45 |

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 M120 arm safe device (see dwg 8880002) with cover removed.

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| 101. | Incorrect spacer thickness behind hex nut | Gage |
| 102. | Incorrect torque of machine screws on timer assembly (6) | Gage |
| 103. | Incorrect torque of machine screws on actuator switch (6) | Gage |
| 104. | Incorrect torque of machine screws on actuator switch assembly (15) | Gage |
| 105. | Incorrect torque of machine screws on actuator switch assembly (15) | Gage |
| 106. | Incorrect torque of machine screws on rods (3) | Gage |
| 107. | Incorrect torque of hex nuts on rods(3) | Gage |
| 108. | Incorrect torque of studs on rods (3) | Gage |
| 109. | Incorrect torque of machine screws (3) | Gage |
| 110. | Incorrect torque of cap screws (6) | Gage |
| 111. | Incorrect torque of machine screws on portable weight (3) | Gage |
| 112. | Incorrect torque of machine screws on bracket (6) | Gage |
| 113. | Glyptal cement missing on specified screwheads | Visual |
| 114. | Any component parts missing, damaged or incorrectly assembled | Visual |
| 115. | Evidence of poor workmanship (see 3.18) | Visual |



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116. O ring not installed.....Visual
117. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor: None defined.

4.3.2.1.2 M120 arm safe device (see dwg 8880002) with cover.

| Categories | Characteristics | Method of Inspection |
|---|---|---|

Critical: None defined.

Major:

101. Mislocation of mounting holes (4) (see dwg. 8880086).....Gage
102. Incorrect diameter of mounting holes (4) (see dwg 8880086).....Gage
103. Incorrect torque of seal screws (3).....Gage
104. Identification plate, missing, incorrect or illegible.....Visual
105. Evidence of poor workmanship (see 3.18).....Visual
106. Mislocation of connector insert pins (see 8880086).....Gage
107. Incorrect depth to connector insert pins (see 8880086).....Gage
108. Improper installation of connector insert (see MS20056053).....Visual/Manual
109. Incorrect parallelism of base surface (see 8880086).....Gage
110. Objective evidence relating to material missing or improper (see 3.1).....Visual
111. Serialization missing or improper (see 3.4).....Visual

Minor: None defined.

4.3.2.1.3 Preparation for delivery (see 5).

| Categories | Characteristics | Method of Inspection |
|---|---|---|

Critical: None defined.



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Major:
101. Preservation, packaging and packing not in accordance with MIL-STD-1169.....Visual
102. Marking not in accordance with MIL-STD-1169.....Visual

Minor:
201. Evidence of poor workmanship (see 3.18).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

| | Test | Test classification |
|---|---|---|
| a. | Circuit resistance | Major |
| b. | Insulation resistance | Major |
| c. | Operation | Major |
| d. | Seal integrity | Major |

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuit resistance, insulation resistance, seal integrity and operation. Each ASD in the lot shall be tested in accordance with 4.4.1 through 4.4.4. If any item fails to comply with 3.6 through 3.9, the item shall be rejected and removed from the lot. At the completion of the operation requirements (3.10) and prior to packing (5.2) each of the three (3) accelerometer switch assemblies in the ASD shall be reset to a safe condition. Circuit A to a shall be tested to show continuity between all three (3) accelerometers in the ASD. The ASD shall be certified and tagged by the government inspector as being in the safe condition at the time of shipment.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8880002). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).



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4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance. The test current shall be 100 ± 1 milliamperes. The measured circuit resistance shall meet the requirements of 3.6 when measured at the pins in accordance with the four wire method of MIL-STD-202, Method 307.

4.4.2 Insulation resistance. The insulation resistance shall meet the requirements of 3.7 in both the safe and armed conditions, when subjected to 500 ± 50 vdc for 10 seconds minimum or 1 minute maximum and tested in accordance with MIL-STD-202, Method 302. The current shall be limited to 5 milliamperes.

4.4.3 Seal integrity. The ASD shall be subjected to a water immersion tube bubble leak test. After the ASD has been soaked for 30 minutes (minimum) in a 15 psig (minimum) atmospheric chamber, the chamber shall be flooded with water sufficient to provide a minimum water depth of one-half inch over the entire ASD. Then the air in the chamber shall be vented to atmospheric pressure, and the ASD observed for a minimum of five minutes for any leakage bubble enemating from the ASD. Test shall meet the requirement of 3.8.

4.4.4 Operation. The ASD shall be mounted in a centrifuge in such a manner that a g-field can be applied suddenly along the X-axis opposite to the direction of the flight (see Figure 1).

4.4.4.1 Temperature shock. Subject the ASD to a temperature of -40°F for a minimum of 4 hours. Prior to removal of ASD from temperature chamber for centrifuge testing, measure and record the circuit resistance, 4.4.1, and the insulation resistance 4.4.2. Within 4 minutes after removing the ASD from the temperature chamber, complete the following tests for Safing, 4.4.4.1.1 and Arming, 4.4.4.1.2, as applicable.

4.4.4.1.1 Safing. With all three accelerometers in the safe position and all the switches connected so that the closure of any normally open switch will be detected and the condition of the monitor circuit observable, apply 11 ± 0.2/0.0 g's to ASD. When the steady-state acceleration is reached, release the swing fixture and maintain the acceleration for a minimum of 10 seconds. No functional switch (normally open) shall close at any time during this portion of the test. Tests shall meet the requirements of 3.9.1.



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4.4.4.1.2 Arming. With all three accelerometers in the safe position apply 14 +0.0, - 0.2 g's to the ASD and release the swing fixture and maintain the acceleration for a minimum of two seconds. The ASD shall arm as evidence by closure of the three arming switches, the opening of the three auxiliary switches. Arming shall occur within a minimum of 0.80 seconds to a maximum of 1.20 seconds. When the g-field is removed, the armed switches shall remain closed. Tests shall meet the requirements of 3.9.2.

4.4.5 Operation. The ASD shall be mounted in a centrifuge in such a manner that the g-field can be applied suddenly along the X-axis, apposite to the direction of flight (see Figure 1).

4.4.5.1 Safing. With all three accelerometers in the safe position and all the switches connected so that the closure of any normally open switch will be detected and the condition of the monitor circuit observable, subject the ASD to 11 +0.2, -0.0 g's for a minimum of 10 seconds. When the steady-state acceleration is applied no functional switch (normally open) shall close at any time during this portion of the test. The test shall meet the requirements of 3.10.1.

4.4.5.2 Arming. Subject the ASD to a temperature of -40°F ± 5°F for a minimum of 1 hour. Within four minutes after removing the ASD from the temperature chamber, complete the following test: With all three accelerometers in the safe position apply 14 +0.0, -0.2 g's to the ASD and release the swing fixture. Maintain the acceleration for a minimum of two seconds. The ASD shall arm, as evidenced by closure of the three arming switches and the opening of the three auxiliary switches. Arming shall occur within a minimum of 0.80 seconds to a maximum of 1.20 seconds. When the g-field is removed, the armed switches shall remain closed. The test shall meet the requirements of 3.10.2.

4.4.5.3 Repeat the test of 4.4.5.2 except that the temperature shall be 160°F ± 5°F. Arming shall occur within a minimum of 0.80 seconds to a maximum of 1.20 seconds.

4.4.5.4 Repeat the test of 4.4.5.2 except that the temperature shall be 80°F. Arming shall occur within a minimum of 0.92 seconds to a maximum of 1.08 seconds.

4.4.6 Mechanical shock.

4.4.6.1 Mechanical shock (-65°F). The unit shall be rigidly mounted to a fixture capable of orientating the unit along each of the three mutually perpendicular axes relative to the shock source (See Figure 1). The unit and fixture shall be secured to an approved shock machine. Prior to performing the shock test the unit shall be conditioned at -65°F ± 5°F in an approved temperature chamber for a minimum of one hour. Following the three shocks in each direction the unit shall be returned to the temperature chamber for a minimum of fifteen minutes before proceeding with the next direction of shock.



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In performing the test the unit shall be removed from the temperature conditioning chamber and the three impacts completed within 5 minutes. Test shall meet the requirements of 3.11.

4.4.6.2 Mechanical Shock. (160°F). The unit shall be rigidly mounted to a fixture capable of orientating the unit along each of the three mutually perpendicular axes relative so the shock source (see Figure 1). The unit and fixture shall be secured to an approved shock machine. Prior to performing the shock test the unit shall be conditioned at 160°F ± 5°F in approved temperature chamber for a minimum of one hour. Following the three shocks in each direction the unit shall be returned to the temperature chamber for a minimum of fifteen minutes before proceduring with the next direction of shock. In performing the test the unit shall be removed from the temperature conditioning chamber and the three impacts completed within 5 minutes. The shall meet the requirements of 3.11.

4.4.7 Operational vibration.

4.4.7.1 Operational vibration. (-40°F). The unit shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes (see Figure 1). The unit shall be vibrated over the frequency ranges of random conditions as defined in 3.12. The ASD shall be conditioned at -40°F ± 5°F in an approved temperature chamber for a minimum of one hour after temperature stabilization (see 6.4), then subjected to the vibration at this temperature according to Table V. After each change of axis, the ASD shall be reconditioned for an additional 15 minutes (minimum) after the chamber temperature has stabilized before vibrating along the new axis. Test shall meet the requirements of 3.12.

4.4.7.2 Operational vibration. (160°F). The unit shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes (see Figure 1). The unit shall be vibrated over the frequency ranges of random conditions as defined in 3.12. The ASD shall be conditioned at 160°F ± 5°F in an approved temperature chamber for a minimum of one hour after temperature stabilization (see 6.4), then subjected to the vibration at this temperature according to Table V. After each change of axis, the ASD shall be reconditioned for an additional 15 minutes (minimum) after the chamber temperature has stabilized before vibrating along the new axis. Test shall meet the requirements of 3.12.

4.4.8 Transportation vibration.

4.4.8.1 Transportation vibration (-65°F). The unit shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes (see Figure 1). The unit and fixture shall be secured to an approved vibration machine and shall be vibrated over the frequency ranges of sinusoidal vibration conditions as defined in 3.13.



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The ASD shall be conditioned at -65°F + 5°F in an approved temperature chamber for a minimum of one hour after temperature stabilization (see 6.4), then subjected to the vibration at this temperature according to Table VI. After each change of axis, the ASD shall be reconditioned for an additional 15 minutes (minimum) after the chamber temperature has stabilized before vibrating along the new axis. Test shall meet requirements of 3.13.

4.4.8.2 Transportation vibration (160°F). The unit shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes (see Figure 1). The unit and fixture shall be secured to an approved vibration machine and shall be vibrated over the frequency ranges of sinusoidal vibration conditions as defined in 3.13. The ASD shall be conditioned at 160°F + 5°F in an approved temperature chamber for a minimum of one hour after temperature stabilization (see 6.4), then subjected to the vibration at this temperature according to Table VI. After each change of axis, the ASD shall be reconditioned for an additional 15 minutes (minimum) after the chamber temperature has stabilized before vibrating along the new axis. Test shall meet requirements of 3.13.

4.4.8.3. Transportation vibration (ambient). The unit shall be ridigly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes (see Figure 1). The unit and fixture shall be secured to an approved vibration machine and shall be vibrated over the frequency range of sinusoidal vibration conditions as defined in 3.13. There shall be 3 vibration sweeps per axis (45 minutes per axis). Test shall meet requirements of 3.13.

4.4.9 Temperature shock. The unit shall be placed in a test chamber at +160°F + 5°F and maintained for 8 + 0.5 hours. The unit will then be removed from the high temperature chamber and, within 5 minutes, placed in a chamber with an internal temperature of -65°F + 5°F and maintained there for 16 + 0.5 hours. Then, within 5 minutes, return the unit to the high temperature chamber. This sequence constitutes one cycle. Three cycles shall be executed. Return the unit to room temperature without forced cooling and test, after a minimum of 4 hours, at ambient temperature (see 6.4). Test shall meet the requirements of 3.14.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The Arm Safe Device shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).



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5.1.2 Level C. The Arm Safe Device shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The Arm Safe Device shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.6,3.7, 3.10.2.1, 3.10.2.2, 3.10.2.3 and 4.3.4.1. Data specified in 3.6, 3.7, 3.10.2.1, 3.10.2.2, 3.10.2.3 and 4.3.4.1 will be delivered on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Physical damage. Physical damage shall include bent or loose pins, split or cracked insulation, damaged connectors, cracks in body.

6.6 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1202A dated 28 September 1972 with Amendment 2 dated 21 June 1973.

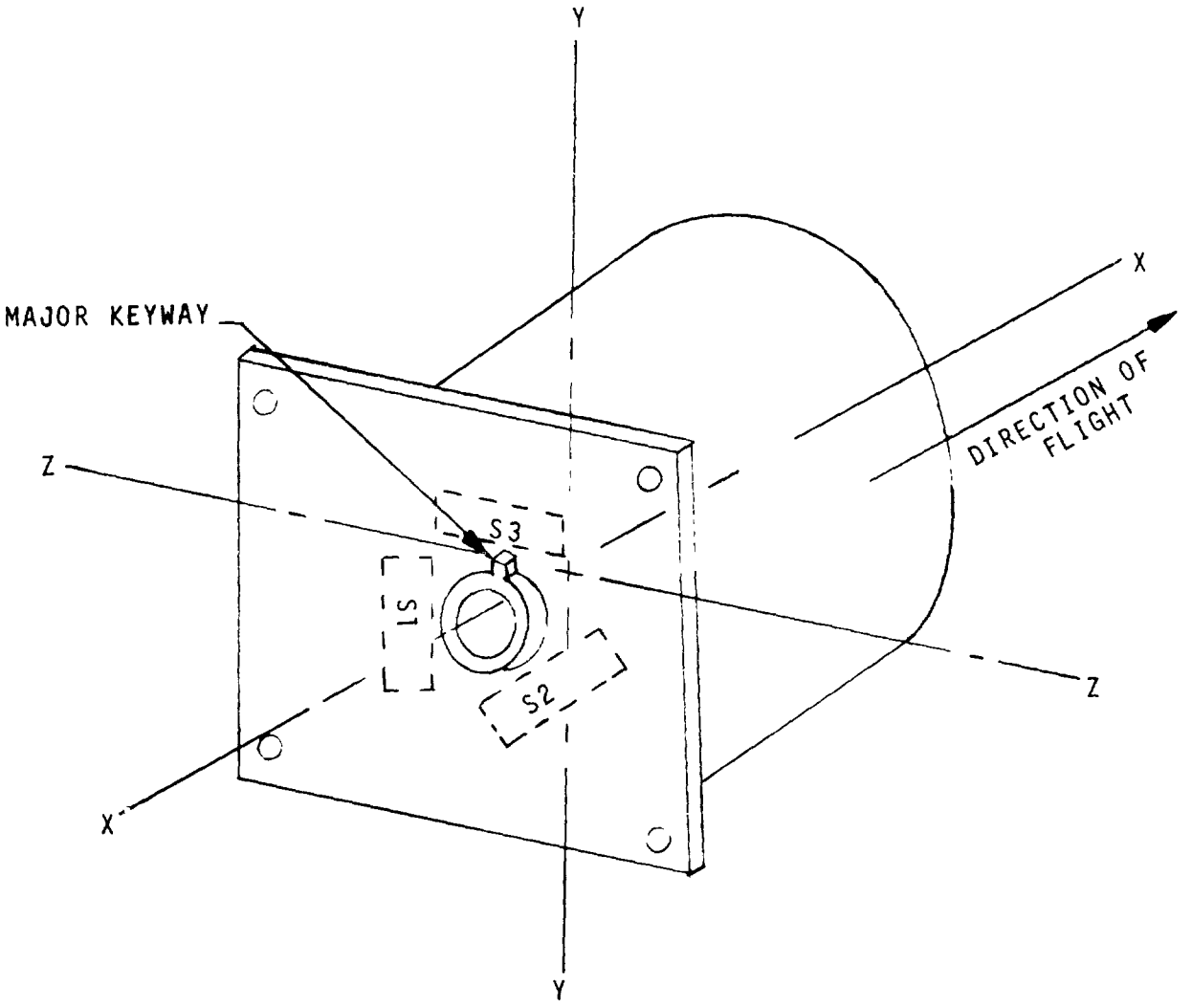
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DEFINITION OF AXES
AND ACCELERATION LOCATION

FIGURE 1



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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